Causal Frameworks for Fair Machine Learning

Counterfactual Fairness and Path-Specific Counterfactual Fairness

A Course Project for

AI4013 / AI3013

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Abstract

Machine learning systems trained on historical data can reproduce and amplify social bias, and conventional statistical fairness criteria often fail to separate genuine causal influence from spurious correlations carried by proxy variables. This report studies how causal inference offers a stronger framework for fair machine learning by explicitly modeling the mechanisms through which sensitive attributes affect predictions. We begin with the foundations of this perspective, including directed acyclic graphs, structural causal models, interventional reasoning through *do*-calculus, and counterfactual analysis. Building on these concepts, we examine counterfactual fairness, mediation analysis, and path-specific effects as increasingly expressive tools for identifying which causal pathways should be blocked and which may be retained.

To connect theory with practice, we reproduce and analyze three representative causal fairness methods on real and simulated datasets. For Counterfactual Fairness, we use Pyro on the Law School Admissions dataset to infer a latent knowledge variable and study the trade-off between fairness and predictive accuracy. For Fair Inference on Outcomes, we apply constrained maximum likelihood estimation with COBYLA to the UCI Adult dataset, restrict discriminatory path-specific effects, and generate fair predictions by marginalizing over mediators. Finally, we implement a Path-Specific Counterfactual Fairness variational autoencoder on the Adult and Berkeley datasets to handle nonlinear relationships using maximum mean discrepancy with random Fourier features. Taken together, these experiments show that causal modeling can reduce discriminatory influence at the individual level while preserving information transmitted through legitimate pathways.

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1. Introduction

The Societal Impact of ML and the Trap of Historical Bias

We are currently witnessing a paradigm shift where machine learning is automating high-stakes decisions across sectors such as lending, insurance, hiring, and criminal justice. A real-world hook for this issue is the COMPAS algorithm, a software used across the United States to predict recidivism. Investigations found it to be systematically biased, assigning higher risk scores to African American defendants compared to white defendants with similar records.

The core question arises: How do we guarantee algorithms are fair when the training data itself might not be? Standard machine learning assumes that training data perfectly represents objective reality. However, the reality is that data is often contaminated by historical prejudices and unequal opportunities. As noted in a 2016 Obama Administration report, technologies can deliberately or inadvertently perpetuate discrimination. If past human decisions were biased, a standard loss function will mathematically force the ML model to replicate and automate that bias in a continuous feedback loop.

Limitations of Existing Statistical Fairness Notions

Several statistical notions of fairness exist, but they suffer from critical limitations:

- **Fairness Through Unawareness (FTU):** The idea is to simply not use protected attributes A (e.g., race, gender) in the model. However, other standard features X often contain discriminatory information analogous to A, acting as proxy variables.
- **Demographic Parity (DP):** Requires equal positive prediction rates across groups, formulated as $P(\hat{Y}|A = 0) = P(\hat{Y}|A = 1)$.
- **Equality of Opportunity (EO):** Requires equal true positive rates, formulated as $P(\hat{Y} = 1|A = 0, Y = 1) = P(\hat{Y} = 1|A = 1, Y = 1)$.

These population-level criteria are based purely on probabilistic independence. They can be mathematically incompatible with one another and, more importantly, they fail to address the underlying causal mechanisms of how unfairness occurs.

Introduction and Formal Definition of Counterfactual Fairness

To address these limitations, we shift to a causal framework. Following Pearl's causal model, let X be the set of observable attributes, U the set of unobservable attributes, Y the outcome, A the protected attributes, and $\hat{Y}$ the predicted outcome. Counterfactual Fairness guarantees fairness at the individual level rather than just the population average. A predictor $\hat{Y}$ is counterfactually fair if, for any individual with features $X = x$ and attribute $A = a$:

$$P(\hat{Y}_{A\leftarrow a}(U) = y|X = x, A = a) = P(\hat{Y}_{A\leftarrow a'}(U) = y|X = x, A = a) \tag{1}$$

Intuitively, an individual should receive the exact same prediction in the real world as they would in a parallel, counterfactual universe where their demographic group was different.

2. Causal Inference Fundamentals

Correlation vs. Causation

A causal effect asks whether changing a specific feature A (the treatment) directly changes the outcome Y. Correlation merely observes that A and Y move together in the data. The gap between association and causation is caused by *confounders*—background variables that cause both the treatment and the outcome, creating a spurious correlation.

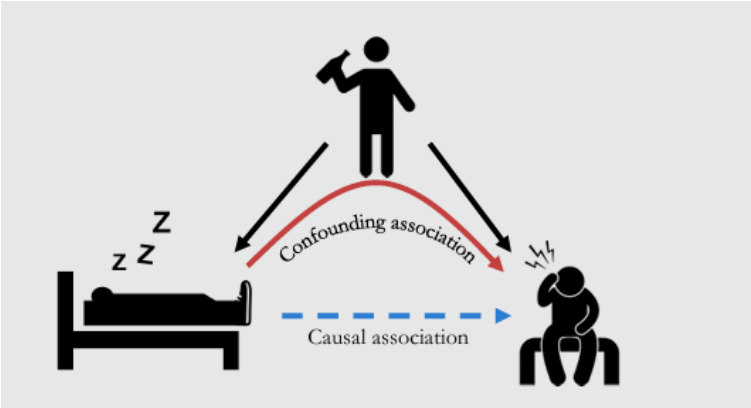


Figure 1: Causation vs. Correlation: Sleeping with shoes is correlated with headaches, but the actual confounder is drinking alcohol.

Notations: DAGs and Structural Causal Equations (SCMs)

We define a causal graph as a Directed Acyclic Graph (DAG), represented by a tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ are the variables and $\mathcal{E}$ are the directed links indicating strict causal relationships. Structural Causal Models (SCMs) power the graph via equations:

$$V_i := f_i(PA_i, U_i) \tag{2}$$

where each variable V_i is determined by its direct parents (PA_i) and unobserved background noise (U_i).

Observational Conditioning vs. Interventional do-calculus

Conditioning, or "seeing," denoted as $P(Y|A = a)$, involves filtering observed data. It leaves the underlying system untouched, meaning confounders still skew the data. Intervening, or "doing," denoted as $P(Y|do(A = a))$, involves actively changing the system to force A to be a. Through graph surgery, the *do*-operator mathematically deletes all incoming edges to the intervened node, effectively breaking the influence of confounders.

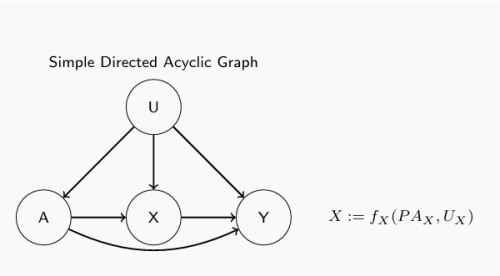


Figure 2: A simple Graphical Causal Model showing the relationships between variables and unobserved noise U.

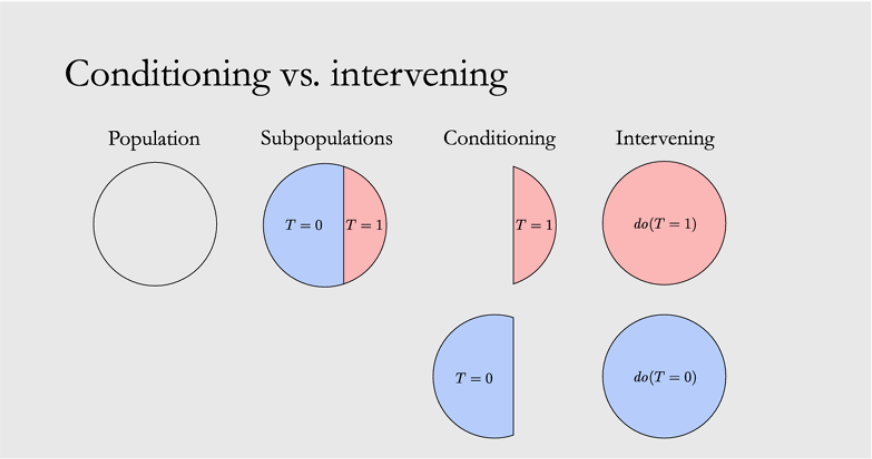


Figure 3: Conditioning subsets the population, while intervening $do(T = 1)$ applies the treatment to the entire population, altering the data generation process.

3. The Mechanics of Counterfactual Fairness

The Fundamental Lemma and Identifiability

Because we cannot simply use standard features X to predict $\hat{Y}$ (as A often causally influences X), we rely on a fundamental lemma: $\hat{Y}$ will be counterfactually fair if it is a function solely of the non-descendants of A.

Translating the theoretical counterfactual into observable data relies on the *Ignorability Assumption*. If there are observed confounders X, we assume $(Y(a), Y(a')) \perp\!\!\!\perp A|X$. The identified equation (Adjustment Formula) becomes $E[Y(a)] = \mathbb{E}_X[E[Y|A = a, X = x]]$. However, if unobserved confounders exist and are missing from our data, the Average Treatment Effect (ATE) becomes completely unidentifiable.

Approximate Fairness: The ϵ-Relaxation

Enforcing strict counterfactual fairness can sometimes severely degrade the predictive accuracy (utility) of the model. To manage this trade-off, we introduce Approximate Fairness (ϵ-Relaxation). Instead of demanding absolute equality, we bound the difference:

$$P(\hat{Y}(a)) \approx P(\hat{Y}(a')) \quad \text{within a tolerance of } \epsilon \tag{3}$$

This mathematical "knob" allows engineers to tune the system, balancing the elimination of bias with the maintenance of a highly accurate classifier.

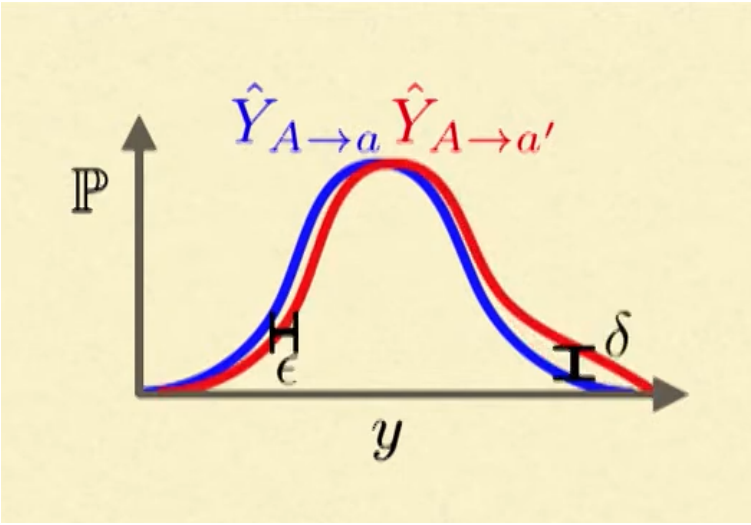


Figure 4: The ϵ-Relaxation bounds the counterfactual distributions, acknowledging the trade-off between strict fairness and utility.

4. Mediation Analysis: Decomposing the Effect

Motivating Example: Why Strict Fairness Fails

Strict Counterfactual Fairness assumes any downstream effect of the sensitive attribute A is unfair. However, consider a firefighter hiring scenario:

- A (Gender) $\to Y$ (Hiring): Direct interviewer bias (Unfair).
- A (Gender) $\to M$ (Physical Strength Test) $\to Y$ (Hiring): A bona fide occupational qualification (Fair).

To satisfy strict counterfactual fairness, we would have to completely ignore the physical test score M because it is causally downstream of gender, thereby destroying valid, legitimate data.

Controlled Direct Effect (CDE) and Nested Counterfactuals

To isolate the unfair direct path, we might use the Controlled Direct Effect (CDE):

$$CDE = E[Y(do(a), do(m))] - E[Y(do(a'), do(m))] \tag{4}$$

The CDE compares two worlds in which the mediator is artificially fixed at the same value m. Any difference in the outcome must therefore travel only through the direct edge $A \to Y$. The drawback is that fixing M performs graph surgery on the arrow $A \to M$, so the indirect path is completely shut off. As a result, the CDE isolates direct discrimination, but it cannot tell us how much of the total effect flows naturally through the mediator.

The key idea in mediation analysis is therefore not to freeze the mediator at an arbitrary constant, but to substitute it with the value it would naturally attain under a counterfactual intervention. This gives the nested potential outcome

$$Y(a, M(a')).$$

It should be read as follows: what would the outcome Y be if we set the sensitive attribute to a, while forcing the mediator to take the value it would have attained had the attribute instead been a'? This construction isolates the direct path from A to Y while still preserving the meaning of the indirect causal mechanism through M.

Identification Assumptions and the Mediation Formula

Let C denote observed pre-treatment confounders for the relationships $A \to M$, $A \to Y$, and $M \to Y$. The attached document highlights four assumptions that make the nested counterfactual identifiable:

1. **No unmeasured confounding for the treatment–outcome link:**

$$Y(a, m) \perp\!\!\!\perp A \mid C.$$

2. **No unmeasured confounding for the treatment–mediator link:**

$$M(a) \perp\!\!\!\perp A \mid C.$$

3. **No unmeasured confounding for the mediator–outcome link:**

$$Y(a, m) \perp\!\!\!\perp M \mid A, C.$$

4. **Cross-world independence:**

$$Y(a, m) \perp\!\!\!\perp M(a') \mid C.$$

In addition, the derivation implicitly uses *consistency* and *positivity*: observed variables agree with their corresponding counterfactuals under realized interventions, and every treatment or mediator level used in the formula has positive probability within each covariate stratum.

Using these assumptions, we can derive the mediation formula step by step. Start from the distribution-level quantity:

$$\begin{aligned}
P(Y(a, M(a')) = y) &= \sum_{c,m} P(Y(a, m) = y \mid M(a') = m, c) \\
&\quad \times P(M(a') = m \mid c)\, P(c) \qquad \text{(law of total probability)} \\
&= \sum_{c,m} P(Y(a, m) = y \mid c) \\
&\quad \times P(M(a') = m \mid c)\, P(c) \qquad \text{(Assumption 4)} \\
&= \sum_{c,m} P(Y = y \mid a, m, c) \\
&\quad \times P(m \mid a', c)\, P(c) \qquad \text{(Assumptions 1–3 + consistency).}
\end{aligned}$$

Taking expectations yields

$$E[Y(a, M(a'))] = \sum_{c,m} E[Y \mid a, m, c]\, P(m \mid a', c)\, P(c). \tag{5}$$

When no confounders are present, this simplifies to

$$E[Y(a, M(a'))] = \sum_{m} E[Y \mid a, m]\, P(m \mid a'). \tag{6}$$

Deriving the Natural Direct and Indirect Effects

With the nested counterfactual identified, the Total Effect (TE), Natural Direct Effect (NDE), and Natural Indirect Effect (NIE) are defined by

$$TE = E[Y(a, M(a))] - E[Y(a', M(a'))] \tag{7}$$

$$NDE = E[Y(a, M(a'))] - E[Y(a', M(a'))] \tag{8}$$

$$NIE = E[Y(a, M(a))] - E[Y(a, M(a'))] \tag{9}$$

The decomposition follows by adding and subtracting the same nested counterfactual term:

$$\begin{aligned} TE &= E[Y(a, M(a))] - E[Y(a', M(a'))] \\ &= \big(E[Y(a, M(a))] - E[Y(a, M(a'))]\big) \\ &\quad + \big(E[Y(a, M(a'))] - E[Y(a', M(a'))]\big) \\ &= NIE + NDE. \end{aligned} \tag{10}$$

This decomposition is exactly what makes mediation useful for fairness. If only the direct path $A \to Y$ is deemed discriminatory, then fairness requires $NDE = 0$. If instead the indirect path $A \to M \to Y$ is the problematic channel, fairness requires $NIE = 0$. Unlike strict counterfactual fairness, mediation analysis can therefore preserve legitimate downstream information rather than discarding every descendant of A.

5. Path-Specific Counterfactual Fairness (The Theory)

The Necessity of Path-Specific Effects (PSE)

Real-world systems have dozens of intertwined variables. We must generalize mediation to arbitrary sets of paths. We classify paths into:

- π (Green Arrows): Interested/Unfair paths we want to evaluate and neutralize.
- π^c (Black Arrows): Baseline/Fair paths we consider legitimate.

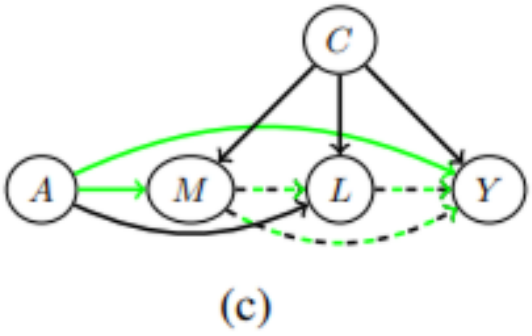


Figure 5: Generalizing to multiple mediators: isolating specific pathways of interest (π) from legitimate baseline paths (π^c).

Deriving the Path-Specific Effect (PSE)

The same logic that produced the Natural Direct Effect (NDE) and Natural Indirect Effect (NIE) extends to general graphs once we partition the causal routes from A to Y into an interested set

π and its complement π^c. In the simple mediation setting, the hybrid counterfactual $Y(a, M(a_0))$ isolates one pathway while holding the other at baseline. For a general graph, we create the analogous hybrid world by sending the intervention value a only through the paths in π, while all remaining paths continue to carry the baseline value a_0.

Let

$$Y_\pi(a, a_0)$$

denote this hybrid counterfactual. For the running graph in Figure 5, the interested paths are the green arrows and the baseline paths are the black arrows, so the hybrid counterfactual becomes

$$Y_\pi(a, a_0) = Y_a(M(a), L(a_0, M(a))). \tag{11}$$

This is the direct analogue of the nested counterfactual used in mediation analysis: we intervene only on the selected unfair routes and leave the remaining routes at their baseline configuration.

The Path-Specific Effect is therefore defined as the contrast between this hybrid world and the fully baseline world:

$$PSE_\pi(a, a_0) = E[Y_\pi(a, a_0)] - E[Y(a_0)]. \tag{12}$$

Substituting the explicit hybrid counterfactual for the current graph gives

$$PSE_\pi(a, a_0) = E[Y_a(M(a), L(a_0, M(a)))] - E[Y(a_0)]. \tag{13}$$

Just as we decomposed the Total Effect into $NDE + NIE$, we can derive the role of the PSE by adding and subtracting the hybrid term:

$$\begin{aligned} TE &= E[Y(a)] - E[Y(a_0)] \\ &= \big(E[Y(a)] - E[Y_\pi(a, a_0)]\big) \\ &\quad + \big(E[Y_\pi(a, a_0)] - E[Y(a_0)]\big) \\ &= \text{remaining effect} + PSE_\pi(a, a_0). \end{aligned} \tag{14}$$

The first term represents the part of the total effect that still travels through the baseline paths, while the second term isolates the contribution of the selected paths in π. Hence, PSE_π is the natural multi-path generalization of mediation effects.

The connection with the earlier quantities is immediate:

- If π contains only the direct path $A \to Y$, then

  $$Y_\pi(a, a_0) = Y(a, M(a_0)),$$

  and therefore

  $$PSE_\pi(a, a_0) = E[Y(a, M(a_0))] - E[Y(a_0, M(a_0))] = NDE. \tag{15}$$

- If π contains only the indirect path $A \to M \to Y$, then

  $$Y_\pi(a, a_0) = Y(a_0, M(a)),$$

  and therefore

  $$PSE_\pi(a, a_0) = E[Y(a_0, M(a))] - E[Y(a_0, M(a_0))] = NIE. \tag{16}$$

This shows why path-specific effects are the correct fairness object in complex systems: instead of classifying all descendants of A as either wholly fair or wholly unfair, they isolate exactly the subset of pathways that a policymaker or model designer wishes to neutralize.

Shpitser's Recursive Rule and Identifiability

To define the counterfactual variable when it receives conflicting signals, we use Shpitser's Recursive Rule. Starting from Y, we walk backward to A:

- For edges not on an unfair path (π^c), set A to the baseline a_0.
- For edges on an unfair path (π), set A to the intervention a.

This results in a final counterfactual of the form $Y_a(M(a), L(a_0, M(a)))$.

Identifiability relies on three ingredients. First, we assume **unconfoundedness**: all common causes C of A, M, L, and Y are observed. Second, we assume the required **cross-world independencies**

$$Y_a(m, l) \perp\!\!\!\perp \{M(a), L(a_0, m)\} \mid C, \qquad L(a_0, m) \perp\!\!\!\perp M(a) \mid C.$$

Third, the **Recanting Witness Criterion** must hold: a path-specific effect is unidentifiable if some node has a child on an interested path and another child on a baseline path. Under these assumptions, together with consistency and positivity, the path-specific distribution is

$$\begin{aligned} P(Y_a(M(a), L(a_0, M(a))) = y) = \int & p(y \mid a, c, m, l)\, p(l \mid a_0, c, m) \\ & \times p(m \mid a, c)\, p(c)\, dc\, dm\, dl. \end{aligned} \tag{17}$$

To derive Eq. (17), start from the nested counterfactual distribution and marginalize over the observed confounders C and the mediators M and L:

$$\begin{aligned} P(Y_a(M(a), L(a_0, M(a))) = y) = \int & P(Y_a(m, l) = y \mid M(a) = m, L(a_0, m) = l, c) \\ & \times P(L(a_0, m) = l \mid M(a) = m, c) \\ & \times P(M(a) = m \mid c)\, p(c)\, dc\, dm\, dl. \end{aligned}$$

This is simply the law of total probability applied to the nested counterfactual world. Next, the cross-world independence assumptions remove the counterfactual conditioning terms:

$$\begin{aligned} P(Y_a(M(a), L(a_0, M(a))) = y) = \int & P(Y_a(m, l) = y \mid c)\, P(L(a_0, m) = l \mid c) \\ & \times P(M(a) = m \mid c)\, p(c)\, dc\, dm\, dl. \end{aligned}$$

Finally, unconfoundedness and consistency let us replace the remaining counterfactual terms by observational quantities:

$$\begin{aligned} P(Y_a(M(a), L(a_0, M(a))) = y) = \int & p(y \mid a, c, m, l)\, p(l \mid a_0, c, m) \\ & \times p(m \mid a, c)\, p(c)\, dc\, dm\, dl, \end{aligned}$$

which is exactly Eq. (17). Consequently, for a real-valued outcome Y, the expectation of the same nested counterfactual is evaluated as

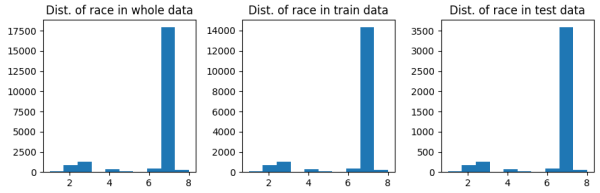
$$E[Y_a(M(a), L(a_0, M(a)))] = \int E[Y \mid a, c, m, l]\, p(l \mid a_0, c, m) \times p(m \mid a, c)\, p(c)\, dc\, dm\, dl. \tag{18}$$

6. Implementation I: Counterfactual Fairness (Kusner et al.)

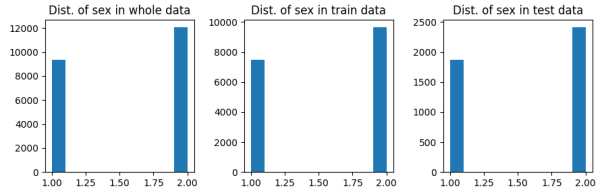
Brief Overview of the Utilized Dataset

To empirically evaluate counterfactual fairness, we used the Law School Admissions Bar Passage dataset. It contains records for approximately 22,000 law school students and includes undergraduate GPA (G), LSAT score (L), race (R), sex (S), and first-year average (FYA, denoted by F). The predictive task is to estimate F from the observed covariates.

In this setting, race (R) and sex (S) are the sensitive attributes. A first inspection of the data shows that both variables are correlated with GPA and LSAT. Those correlations should not be interpreted as direct evidence of innate ability; they can also arise from unequal access to schooling, coaching, and broader social opportunity. Because G and L are descendants of R and S in the causal graph, they act as proxies for sensitive information. As a result, simply removing R and S from the feature vector does not eliminate the discriminatory signal.



(a) Sex distribution across the train–test split, showing preserved balance.



(b) Race distribution across the split used in the experiments.

Figure 6: Dataset split diagnostics for the Law School Admissions data.

Computational Methodology and Sampling Techniques

Kusner et al. propose a structural causal model in which a latent "Knowledge" variable K represents the student's underlying academic preparedness. The key modeling choice is that K causally affects GPA, LSAT, and FYA, but is not itself a descendant of race or sex. By the fundamental lemma of counterfactual fairness, any predictor built solely from such non-descendants can satisfy the fairness criterion.

We implemented the generative model in Pyro. The structural equations are

$$K \sim \mathcal{N}(0, 1), \tag{19}$$

$$G = w_{KG}K + w_{RG}R + w_{SG}S + \epsilon_G, \tag{20}$$

$$L = w_{KL}K + w_{RL}R + w_{SL}S + \epsilon_L, \tag{21}$$

$$F = w_{KF}K + \epsilon_F. \tag{22}$$

Because K is latent, it must be inferred from the observed variables. We conditioned the model on (G, L, F, R, S) for each training instance and used importance sampling with 10 posterior samples per individual to approximate $p(K \mid G, L, F, R, S)$. The posterior mean of K was then used as a point estimate for downstream prediction.

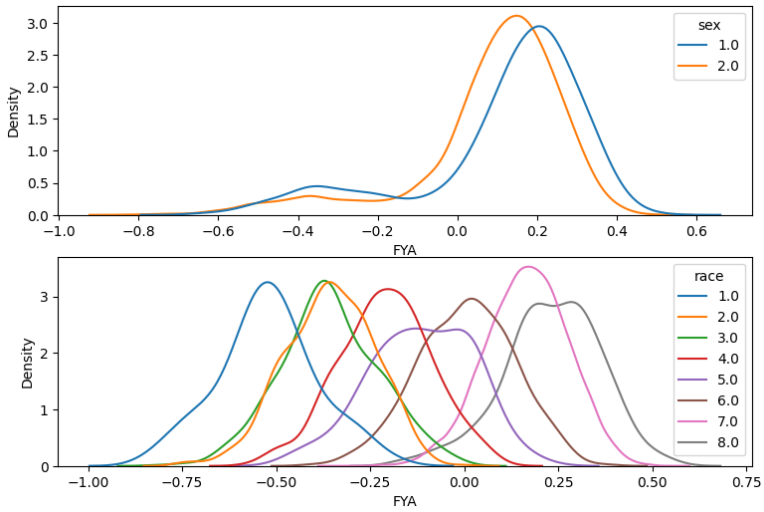


Figure 7: Empirical density of the inferred latent knowledge variable K compared against a standard Gaussian prior. The overlap provides a basic sanity check for posterior inference.

To assess the utility–fairness trade-off, we trained four linear regression baselines using the Adam optimizer (learning rate 0.05) and mean squared error loss:

1. **Full model:** uses all observed variables (R, S, G, L) and acts as the explicitly biased accuracy upper bound.
2. **Unaware model:** uses only (G, L) and therefore removes protected attributes but retains proxy bias.
3. **Fair-K model:** predicts FYA from the inferred latent variable K alone.
4. **Fair-Add model:** predicts from residualized versions of GPA and LSAT after regressing out the part explained by R and S.

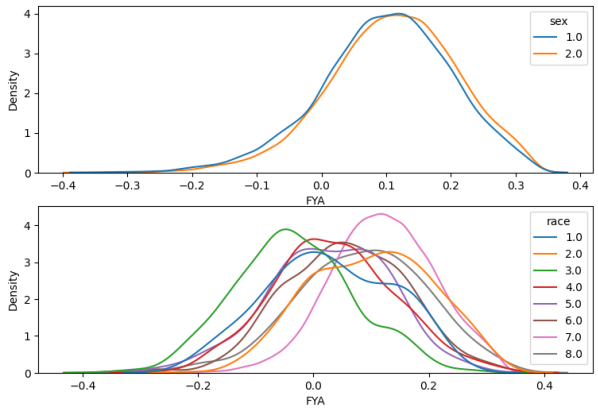
Results and Evaluation on the Dataset

Model performance was evaluated on a stratified 20% test split. Table 1 summarizes the resulting RMSE values.

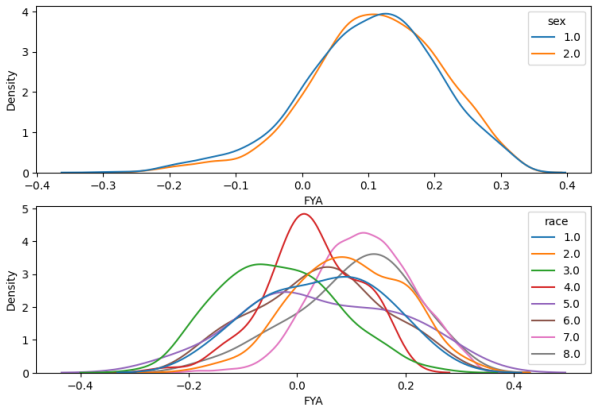
To make the fairness behavior visible, we plotted the kernel density estimates of predicted FYA for different demographic groups. A fair model should show substantially more overlap between group-conditional prediction distributions than an unfair one.

Model	RMSE	Counterfactually Fair?
Full	0.615	No (×)
Unaware	0.671	No (proxy bias remains)
Fair Add	0.723	Yes (✓)
Fair K	0.746	Yes (✓)

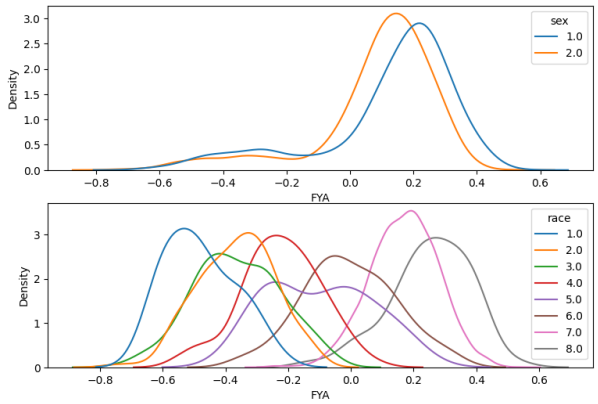
Table 1: Predictive accuracy and fairness guarantees for the four models on the Law School dataset.



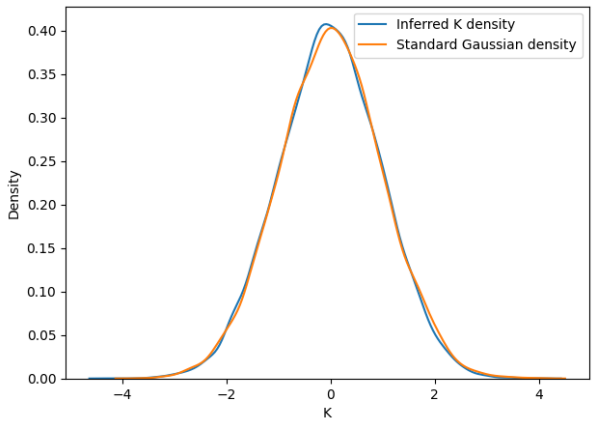
(a) Unaware model: group separation persists because proxy information leaks through GPA and LSAT.



(b) Full model: the strongest separation appears when the model directly exploits sensitive information.



(c) Fair-K model: the distributions move much closer once prediction depends only on the non-descendant latent variable.



(d) Fair-Add model: residualization also reduces separation while retaining more predictive signal than the latent-only baseline.

Figure 8: Group-conditional prediction densities for the four counterfactual fairness baselines.

Discussion of Results: Paving the Way for Path-Specific Fairness

The results clearly expose an accuracy–fairness trade-off. The full model achieves the lowest RMSE because it benefits from historical signal embedded in the demographic pathways, but precisely that signal makes the predictor unfair. The unaware model performs only slightly worse, which confirms that dropping the protected variable is not enough once descendants of that variable remain in the feature set.

The two causally motivated models remove that dependency, but at a measurable cost in predictive performance. The fair-K model is conceptually clean because it depends only on a non-descendant latent variable, yet it suffers the largest utility drop. The fair-add model recovers some accuracy by retaining residualized components of GPA and LSAT, but this gain comes from stronger modeling assumptions about how the unfair portion can be subtracted away. In more complex nonlinear settings, those linear residual corrections become fragile. That observation motivates the move from strict counterfactual fairness to path-specific counterfactual fairness.

7. Implementation II: Fair Inference on Outcomes (Nabi & Shpitser)

Transition to Path-Specific Effects

The previous section treated all descendants of the sensitive attribute as potentially contaminated. Nabi and Shpitser refine that perspective by distinguishing between causal pathways that are normatively unacceptable and pathways that may legitimately transmit predictive information. Their framework therefore targets unfairness at the level of path-specific effects rather than at the level of entire variables.

Causal Formulation and the Adult Dataset

We evaluated this framework on the UCI Adult dataset, where the prediction task is whether a person's income exceeds \$50K. Gender is treated as the sensitive attribute A, while marital status (M), education (L), and work-related variables (R) act as mediators between A and the outcome Y.

The fairness judgment used in this experiment declares the direct path $A \to Y$ and all paths that pass through marital status to be disallowed. Paths that move through education or work variables without relying on the disallowed marital-status mechanism are treated as permissible. Under logistic and linear submodels, the corresponding path-specific effect on the odds-ratio scale can be written as

$$\mathrm{PSE} = \exp\left(\theta_A^Y + \theta_M^Y \theta_A^M + \theta_L^Y \theta_M^L \theta_A^M + \sum_i \theta_{R_i}^Y \left(\theta_M^{R_i} \theta_A^M + \theta_L^{R_i} \theta_M^L \theta_A^M\right)\right). \tag{23}$$

Perfect repair along those paths corresponds to $\mathrm{PSE} = 1$.

Constrained Optimization and Fair Prediction

The central algorithmic step is to solve a constrained maximum-likelihood problem. Instead of fitting the empirical distribution directly, we seek a repaired distribution p^* that stays close to the observed data while forcing the targeted PSE into a fairness band of $[0.95, 1.05]$. We implemented this constrained estimation using COBYLA.

Prediction at test time requires more than plugging repaired parameters into a standard logistic classifier. Test examples are still drawn from the original unfair world, so the model must marginalize over repaired mediator distributions. We therefore compute fair predictions by sampling from $p^*(M, L, R \mid C)$ and averaging the resulting outcome probabilities, yielding a Monte Carlo estimate of $\mathbb{E}^*[Y \mid W]$.

Results and Visual Diagnostics

Table 2 summarizes the main outcome of the constrained inference procedure.

Model	**Accuracy**	**PSE (odds-ratio)**
Unconstrained (standard MLE)	81.8%	3.54 (discriminatory)
Drop-A baseline	81.8%	High (proxy bias remains)
Constrained KL-fair model	73.6%	1.05 (within fairness band)

Table 2: Predictive accuracy and path-specific fairness on the Adult test set.

The drop-A baseline again demonstrates the failure of fairness through unawareness: removing gender does not meaningfully change the predictive accuracy because the descendants of gender continue to carry the same discriminatory information. By contrast, the constrained model succeeds in reducing the targeted PSE from 3.54 to 1.05, but that repair lowers accuracy to 73.6%.

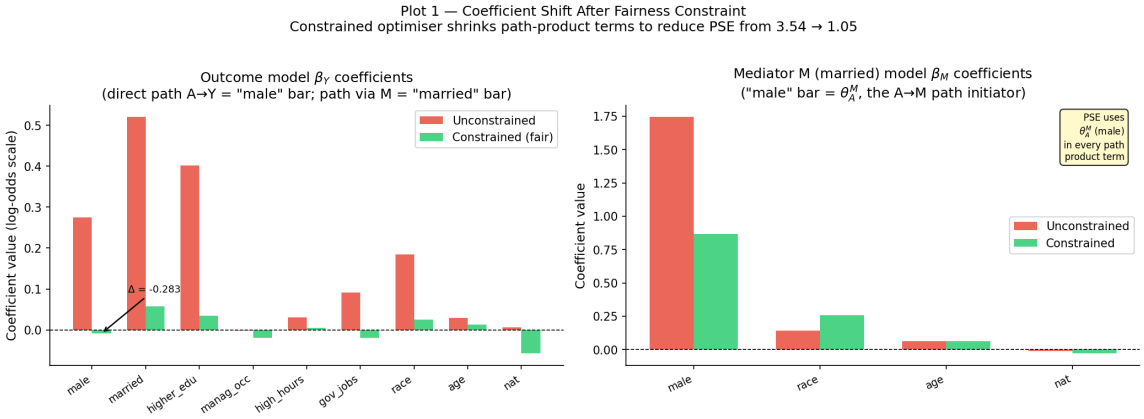


Figure 9: Coefficient shift under constrained optimization. The largest shrinkage occurs along the direct $A \to Y$ effect and the path-initiating $A \to M$ term, which are precisely the terms driving the disallowed pathways.

The distributional plots make the repair visible at the individual-prediction level, but they do not fully capture the loss of predictive power across all decision thresholds. For that reason, we also examined the ROC curves of the three models.

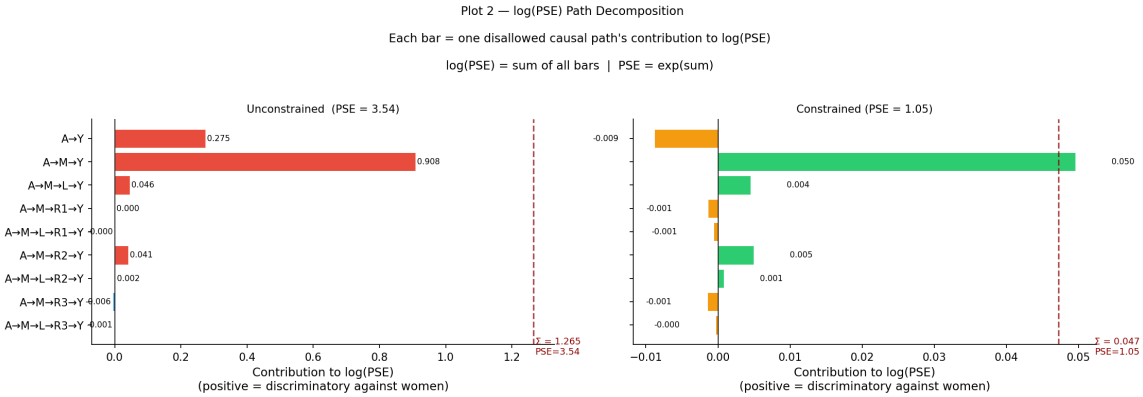


Figure 10: Path decomposition of log(PSE). Before repair, the direct path and the paths through marital status dominate the unfair effect. After projection into the feasible set, the disallowed contributions are sharply compressed.

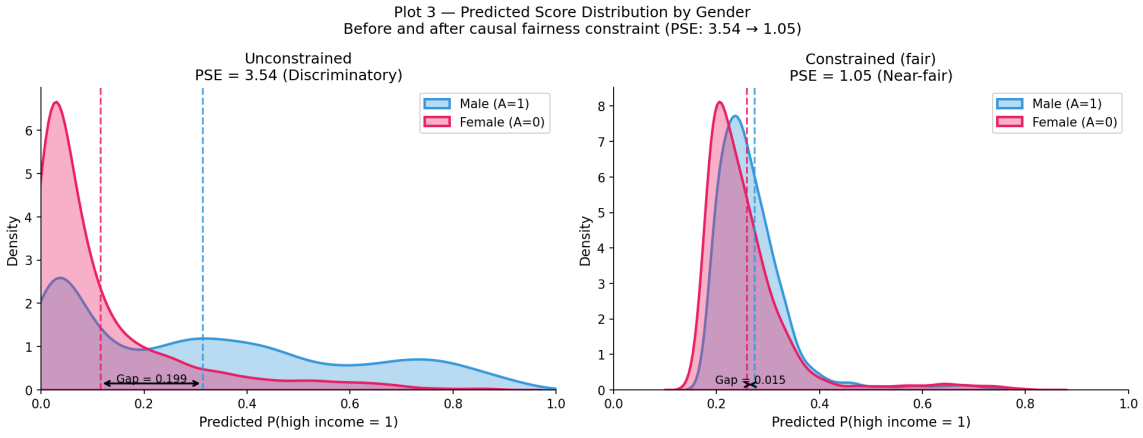


Figure 11: Predicted score distributions by gender. The constrained model narrows the gap between male and female score distributions by integrating over the repaired mediator distribution.

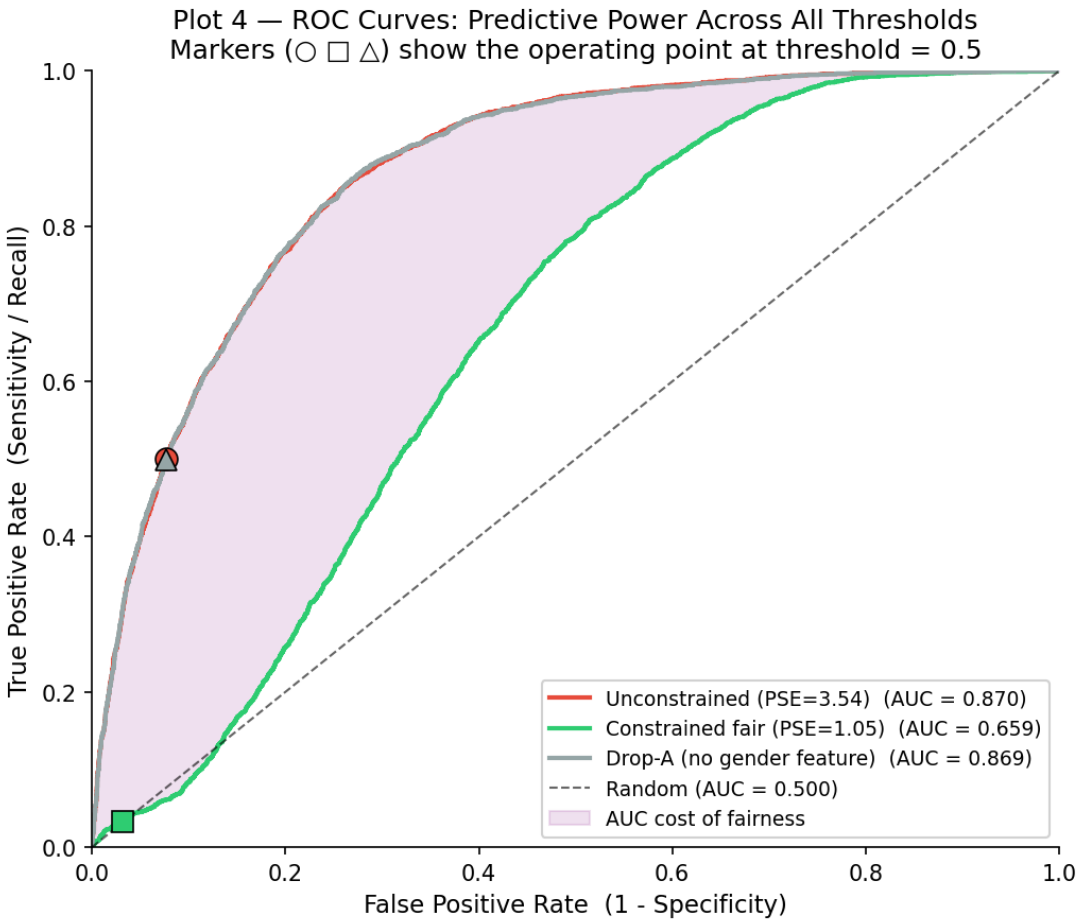


Figure 12: ROC comparison across thresholds. The constrained fair model incurs a clear AUC penalty relative to the unconstrained and drop-A baselines, which quantifies the utility cost of enforcing the path-specific constraint.

The same trade-off is summarized from a different perspective in the approximate Pareto frontier shown in Figure 13. As the model is pushed closer to perfect fairness (PSE $\to$ 1), the achievable test accuracy declines.

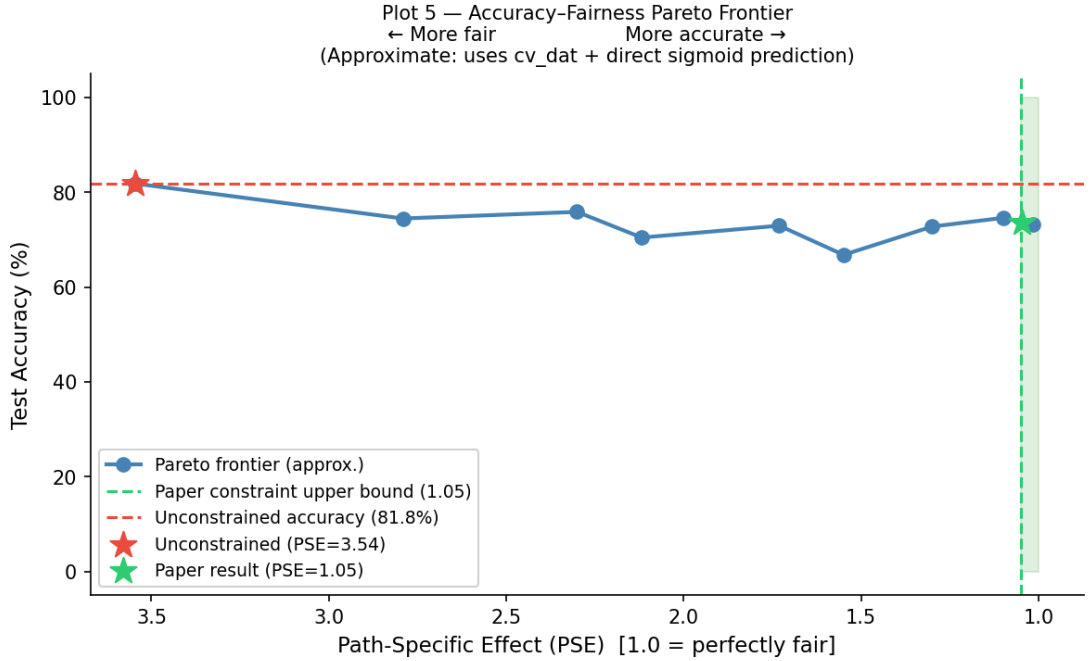


Figure 13: Approximate accuracy–fairness Pareto frontier. The paper's constrained solution sits near the fairness boundary, illustrating the price of reducing the disallowed path-specific effect.

This parameter-constrained approach is effective when the structural model is sufficiently expressive and the fairness judgment can be written directly as restrictions on path products. However, the repair remains global: it acts by changing model parameters rather than by reconstructing an individual's descendants in a fair counterfactual world. That limitation motivates the final implementation.

8. Implementation III: Path-Specific Counterfactual Fairness (PSCF)

Why Linear Parameter Corrections Become Insufficient

The previous implementations assume that linear structural equations are expressive enough to separate fair and unfair information. In practice, that assumption can be too rigid. Real datasets often contain nonlinear interactions between the sensitive attribute, legitimate covariates, and downstream mediators. When a linear structural equation model cannot capture those interactions, they are absorbed into residual noise terms. The result is subtle but important: the residuals become correlated with the sensitive attribute, and fairness can leak back into the reconstructed variables.

Path-Specific Counterfactual Fairness addresses that failure mode by moving from explicit linear residualization to latent-variable modeling. Instead of subtracting the unfair component algebraically, PSCF learns a flexible latent representation that preserves individual-specific information while

directly penalizing dependence on the sensitive attribute.

PSCF-VAE Architecture: Explicit Latents and MMD

Following Chiappa and Gillam, each descendant variable that requires repair is assigned its own latent variable. For a mediator M, the linear equation

$$M = \theta^m + \theta^m_A A + \theta^m_C C + \epsilon_m \tag{24}$$

is replaced by the generative model

$$M \sim p_\theta(M \mid A, C, H_M). \tag{25}$$

The key design principle is that H_M should carry only fair, individual-specific information. To enforce that requirement, the model penalizes any dependence between H_M and the sensitive attribute using the maximum mean discrepancy objective

$$\mathrm{MMD}\big[\tilde{p}(H_M \mid A = 0),\, \tilde{p}(H_M \mid A = 1)\big] \to 0. \tag{26}$$

In other words, the latent representation is encouraged to satisfy $H_M \perp\!\!\!\perp A$. This allows nonlinear structure to be captured in the latent space while preventing the latent variables from preserving the disallowed demographic signal.

Because the exact posterior $p_\theta(H \mid V)$ is intractable, PSCF uses variational inference. An encoder $q_\phi(H \mid V^*)$ and decoder $p_\theta(V \mid A, C, H)$ are trained jointly to optimize the evidence lower bound together with the MMD penalty. The reparameterization trick then enables end-to-end gradient-based training.

Experimental Setup on the Adult Dataset

We implemented PSCF-VAE on the Adult dataset, introducing three latent correction variables: H_M for marital status, H_L for education, and H_R for work-related descendants. The causal graph used for this family of experiments is shown in Figure 14; the left panel corresponds to the Adult dataset, while the right panel provides the companion German Credit reference graph used in the broader experimental pipeline.

Training followed a two-stage schedule for the MMD weight β. During the first 5,000 steps, we set $\beta = 0$ so that the VAE could first learn a stable reconstruction of the data distribution. After that warm-up stage, we increased the penalty to $\beta = 1000$ in order to actively enforce the independence constraint between the latent variables and the sensitive attribute.

Steps	**Biased Acc.**	**Fair Acc.**	**MMD** H_M	**MMD** H_L	**MMD** H_R
5,000	0.8248	0.7982	1134.24	6.74	0.41
10,000	0.8259	0.7752	0.11	0.00	0.00
15,000	0.8301	0.7688	0.01	0.00	0.00

Table 3: PSCF-VAE results on the Adult dataset. The MMD columns are reported after scaling by 10^4.

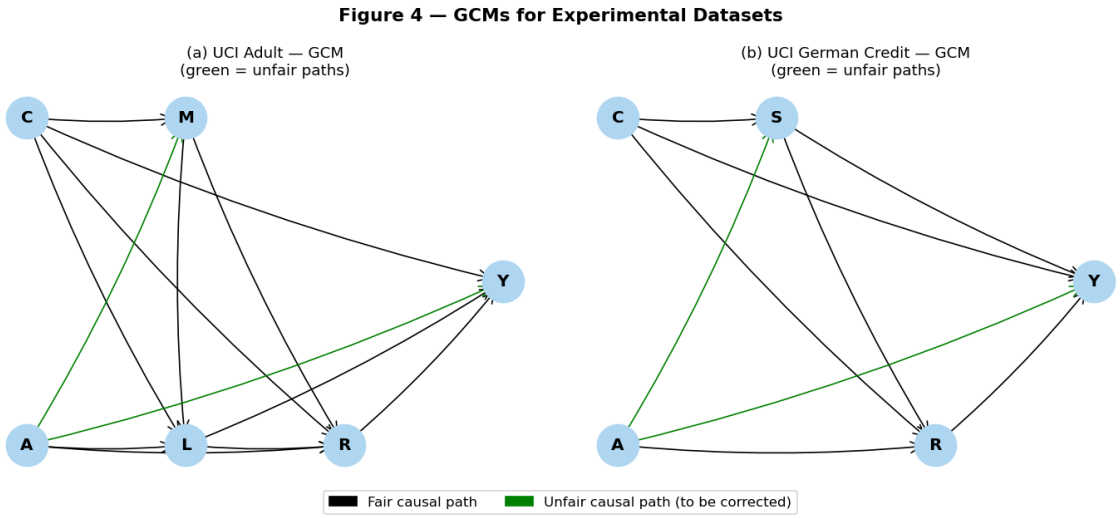
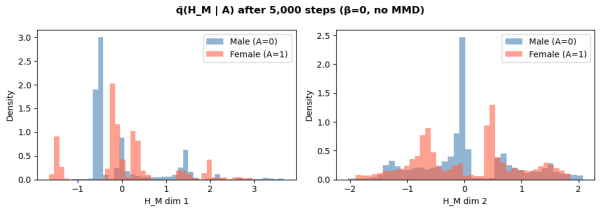


Figure 14: Graphical causal models used in the PSCF experiments. Green edges denote unfair paths to be repaired. The Adult graph appears on the left and the German Credit reference graph on the right.

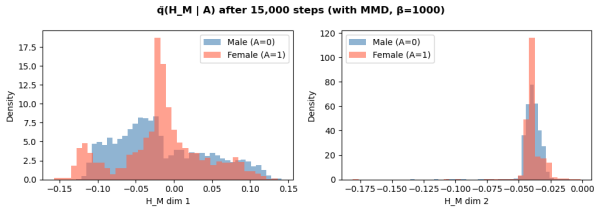
Results and Diagnostic Interpretation

Table 3 captures the central behavior of PSCF-VAE. Once the MMD penalty is activated, the dependence of the marital-status latent H_M on the sensitive attribute collapses dramatically, while fair predictive accuracy degrades only moderately. This is precisely the trade-off PSCF is designed to achieve: it preserves more individual-specific information than rigid linear residualization, but suppresses the demographic component inside the latent space.

That mechanism becomes directly visible in the empirical latent posteriors shown in Figure 15. Before the MMD term is switched on, the male and female posteriors of H_M are visibly separated. After regularization, the two distributions overlap much more closely, indicating that the sensitive information has been substantially removed from the latent code.



(a) Adult latent posterior at 5,000 steps with $\beta = 0$. Clear separation remains between the two groups.



(b) Adult latent posterior at 15,000 steps with $\beta = 1000$. The overlap increases once MMD regularization is active.

Figure 15: Latent posterior diagnostics for the marital-status latent variable H_M on the Adult dataset.

The broader training dynamics are shown in Figures 16 and 17. The left panels correspond to the Adult experiment discussed here, while the right panels provide the analogous German Credit reference run. In both cases, the turning point coincides with the activation of the MMD regularizer.

At test time, PSCF makes a fair prediction by first inferring the individual's latent state and then reconstructing the descendants under the baseline attribute value a_0. Conceptually, the model asks: "What would this person's mediators look like in a counterfactual world where the unfair pathway is turned off, while keeping the person's fair latent characteristics unchanged?" That individual-level

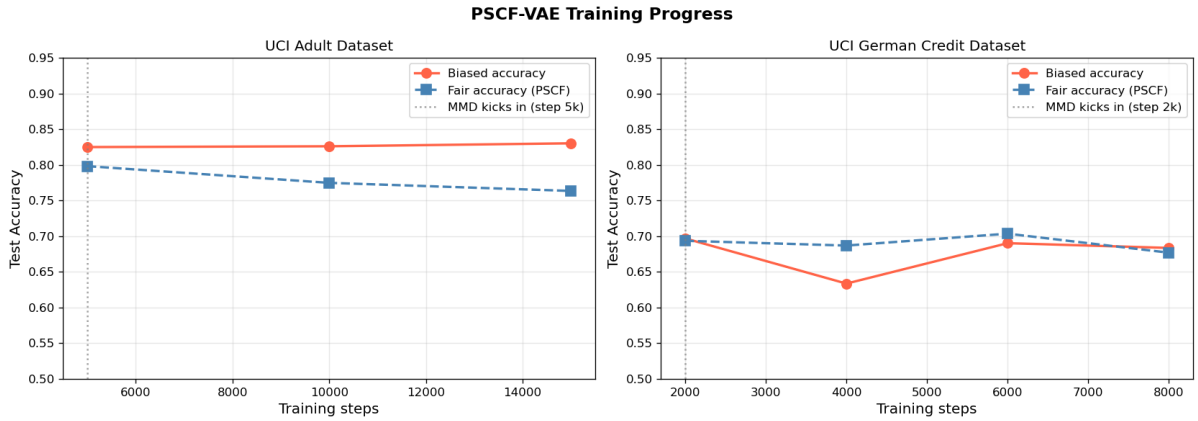


Figure 16: Training accuracy trajectories for PSCF-VAE. When the MMD penalty is activated, fair accuracy drops slightly, but remains substantially usable while the biased predictor stays nearly unchanged.

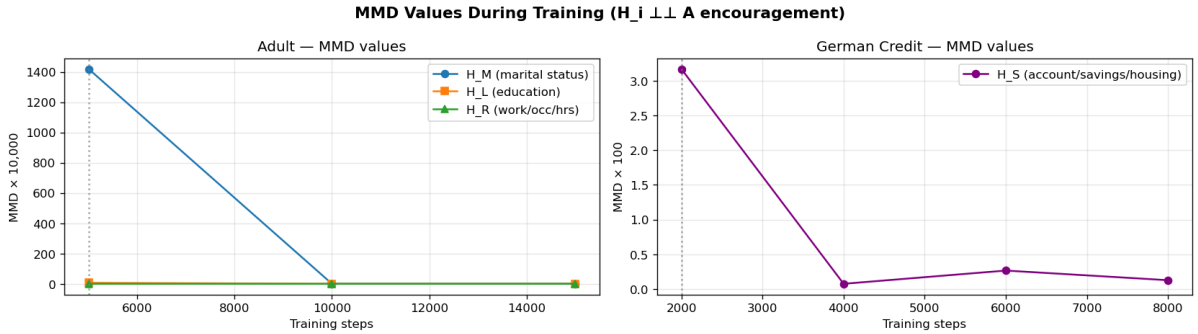


Figure 17: MMD values during PSCF-VAE training. The sharp decline after the warm-up stage shows that the latent variables become substantially less dependent on the sensitive attribute once regularization is enforced.

repair is the main conceptual advantage of PSCF over parameter-only constraints.

Overall, the three implementations in this report form a clear progression. Counterfactual fairness removes sensitive information by restricting predictors to non-descendants; fair inference on outcomes constrains only the disallowed causal paths; and PSCF finally generalizes the correction to nonlinear settings by learning fair latent representations. Together, they show how increasingly expressive causal models let us separate useful predictive information from discriminatory influence with greater precision.